Comprehensive System Analysis & Methodology Report

1. System Overview: The Dual-Branch Architecture

This system utilizes a **Dual-Branch Deep Learning Model** to detect malware by analyzing two distinct modalities: **Spatial Structure** (visual representation) and **Temporal Behavior** (execution sequence). This design aligns with the core goal of capturing complementary features that single-modal models miss.

1.1 Architectural Components

A. Spatial Branch (CNN) - The "Eye" of the System

Defined in `models/cnn_branch.py`, this branch treats malware binaries as grayscale images to detect structural anomalies. * **Input**: (Batch, 1, 64, 64) grayscale images. * **Layer Structure (Deep Feature Extraction)**: 1. **Conv Block 1**: 32 filters (3x3), BatchNorm, ReLU, MaxPool (2x2). 2. **Conv Block 2**: 64 filters (3x3), BatchNorm, ReLU, MaxPool (2x2). 3. **Conv Block 3**: 128 filters (3x3), BatchNorm, ReLU, MaxPool (2x2). 4. **Conv Block 4**: 256 filters (3x3), BatchNorm, ReLU, MaxPool (2x2). * **Feature Aggregation**: * **Adaptive Pooling**: `AdaptiveAvgPool2d((4, 4))` ensures the model can handle variable input sizes if needed, flattening them to a fixed vector. * **Projection**: Flattened vector $\rightarrow$ Fully Connected (512) $\rightarrow$ Dropout (0.5) $\rightarrow$ Output Feature Vector (256-dim).

B. Temporal Branch (BiLSTM) - The "Brain" of the System

Defined in `models/lstm_branch.py`, this branch analyzes the *sequence* of operations (API calls) to understand intent. * **Input**: Sequence of integers (API call IDs). * **Embedding Layer**: Converts discrete API IDs into dense vectors (Default: 128-dim). * **Recurrent Core**: * **Bidirectional LSTM**: 2 stacked layers with 128 hidden units. "Bidirectional" means it reads the sequence forwards and backwards, capturing context from both past and future API calls. * **Attention Mechanism (Crucial Feature)**: * Instead of just taking the last hidden state, the model performs a "Self-Attention" step. * It calculates a "relevance score" for every time step: $Score_t = Linear(H_t)$. * It computes weights via Softmax: $\alpha_t = \frac{exp(Score_t)}{\sum exp(Score)}$ * **Context Vector**: The final representation is a weighted sum $\sum \alpha_t H_t$. This allows the model to "focus" on specific malicious API calls (e.g., `CreateRemoteThread`) while ignoring benign setup code. * **Projection**: Context Vector $\rightarrow$ Fully Connected (512) $\rightarrow$ Dropout (0.5) $\rightarrow$ Output Feature Vector (256-dim).

C. Fusion & Classification Layer

- **Fusion**: The 256-dim vectors from CNN and LSTM are fused (concatenated to 512-dim).
  Classifier Head:
  - Linear (512 $\rightarrow$ 512) $\rightarrow$ ReLU $\rightarrow$ Dropout (0.3).
  - Linear (512 $\rightarrow$ 256) $\rightarrow$ ReLU $\rightarrow$ Dropout (0.3).
  - Linear (256 $\rightarrow$ 2) [Benign/Malware].

1.2 Addressing Class Imbalance

The dataset contains a significant imbalance (Malware >>> Benign). The training pipeline (`train.py`) employs a two-pronged strategy to prevent model bias: 1. **Weighted Random Sampling**: Forces every training batch to contain a balanced 50/50 mix of malware and benign samples, preventing the model from ignoring the minority class. 2. **Focal Loss**: Replaces standard Cross-Entropy loss. It applies a modulating factor $(1 - p_t)^\gamma$ to down-weight "easy" examples, forcing the model to focus on hard-to-classify samples (edge cases).

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2. Feature Fusion Strategies: Methodology & Performance

The system supports three strategies to bridge the CNN and LSTM branches.

2.1 Strategy Definitions

1. **Concatenation (`concat`)**: Joins feature vectors end-to-end ($[F_{cnn}, F_{lstm}]$). Preserves all information.
2. **Addition (`add`)**: Adds feature vectors element-wise. Forces a shared semantic space but compresses information.
3. **Attention (`attention`)**: Learns dynamic weights to emphasize the more reliable branch for each specific sample (e.g., trusting LSTM more if the CNN image is noisy).

2.2 Performance Comparison

Analysis based on `visualizations/fusion_comparison_table.png`

Fusion Method	Accuracy	Recall (Malware)	Specificity (Benign)	Analysis
Concat	**98.25%**	**98.38%**	92.25%	**Best Overall**. By keeping all features available, the classifier finds the optimal decision boundary. It is the most reliable for detection metrics.

Attention	97.74%	97.83%	**93.66%**	**Best Specificity**. It excels at reducing false alarms by intelligently filtering out noisy features, making it ideal for environments where False Positives are costly.
Add	97.66%	97.81%	90.85%	**Underperformer**. The information loss from "squashing" two distinct modalities into one sum proved detrimental to accuracy.

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3. Adversarial Defense: Mutation Strategies

To ensure robustness (Thesis Objective 2), the model is trained against "Mutated" samples generated by the `ThreatMutator` module.

3.1 Strategy Definitions

1. **Random Insert**: Injects random API calls ("noise") into the sequence. Simulates **Dead Code Insertion** to disrupt N-gram signatures.
2. **Random Replace**: Overwrites valid API calls with random ones. Simulates **Polymorphism** and instruction substitution.
3. **Noise Append**: Adds junk data to the end of the file. Simulates **Overlay Attacks** or wrapper injections.

3.2 Comparative Robustness Analysis

Analysis based on `experiments/robustness_*.json`

We evaluated which training strategy provided the best "Universal Defense" (Cross-Protection).

Training Strategy	vs. Insert Attack	vs. Replace Attack	vs. Append Attack	Verdict

Insert Strategy	**92.4%**	**92.6%**	96.8%	**The Winner**. It learned robust long-term dependencies. Remarkably, it defended against "Replace" attacks better than the "Replace" model itself.
Append Strategy	91.0%	84.0%	**94.0%**	**Specialist**. Great at ignoring tail noise, but failed to generalize to internal structural changes (Replace/Insert).
Replace Strategy	62.4%	60.0%	90.8%	**Failure**. Training on "Replacement" data (destroying valid signals) confused the model, leading to catastrophic failure. It typically hurts more than it helps.

Key Insight: **Random Insert** is the superior defense strategy. It forces the LSTM to be flexible with sequence interruptions, effectively handling both insertions and content corruption.

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4. Final Model Performance (Concat Strategy)

Analysis based on `checkpoints/evaluation_results.json`

The final deployed model (using **Concatenation**) demonstrates production-ready capabilities.

4.1 Confusion Matrix Breakdown

	Predicted Benign	Predicted Malware

Actual Benign	**126 (TN)**	16 (FP)
Actual Malware	72 (FN)	**6282 (TP)**

Malware Detection:

- **Precision (99.75%)**: Highly trustworthy. If it alerts, it is almost certainly a threat.
- **Recall (98.87%)**: Missed only 72 out of 6354 samples.

Benign Handling:

- **Precision (63.64%)**: Statistically low due to the small benign dataset size, but functionally acceptable (only 16 false claims).
- **Recall (88.73%)**: The model errs on the side of caution (Paranoia), preferring to flag a benign file rather than miss a malware file.

4.2 Conclusion

The system successfully meets the thesis objectives: 1. **Dual-Branch Fusion**: Validated as superior to single-method approaches (98.65% Accuracy). 2. **Adversarial Defense**: Validated "Random Insert" as a robust defense mechanism against evasion. 3. **Operational Viability**: Confirmed by high True Positive rate (6282) and low False Positive rate (16).

Appendix A: Technology Stack Summary

C	Technology	Version	Purpose
Core Framework	Python	3.9+	Primary programming language
	PyTorch	2.1+	Deep Learning backend (AutoGrad, Neural Nets)
	Torchvision	0.16+	Image transformations and datasets
Data Processing	Numpy	1.24+	Tensor numerical operations
	Pandas	2.0+	Dataframe manipulation for CSV/Results

	Pillow	10.0+	Image processing (Binary $\to$ Image conversion)
ML Ecosystem	Scikit-Learn	1.3+	Metrics calculation (Recall, Precision, F1)
	Tqdm	4.65+	Training progress bars
Visualization	Matplotlib	3.7+	Plotting graphs and tables
	Seaborn	0.12+	Heatmaps (Confusion Matrices)
DevOps & CI/CD	GitHub Actions	-	Automated pipeline orchestration
	Ubuntu	Latest	CI runner environment
	xhtml2pdf	-	Report generation utility